Stepwise Benchmarking in the Data Scarce Setting of Groundwater Heat Plume Modeling

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Collaborators:
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Alexander Heinlein (TU Delft, the Netherlands)

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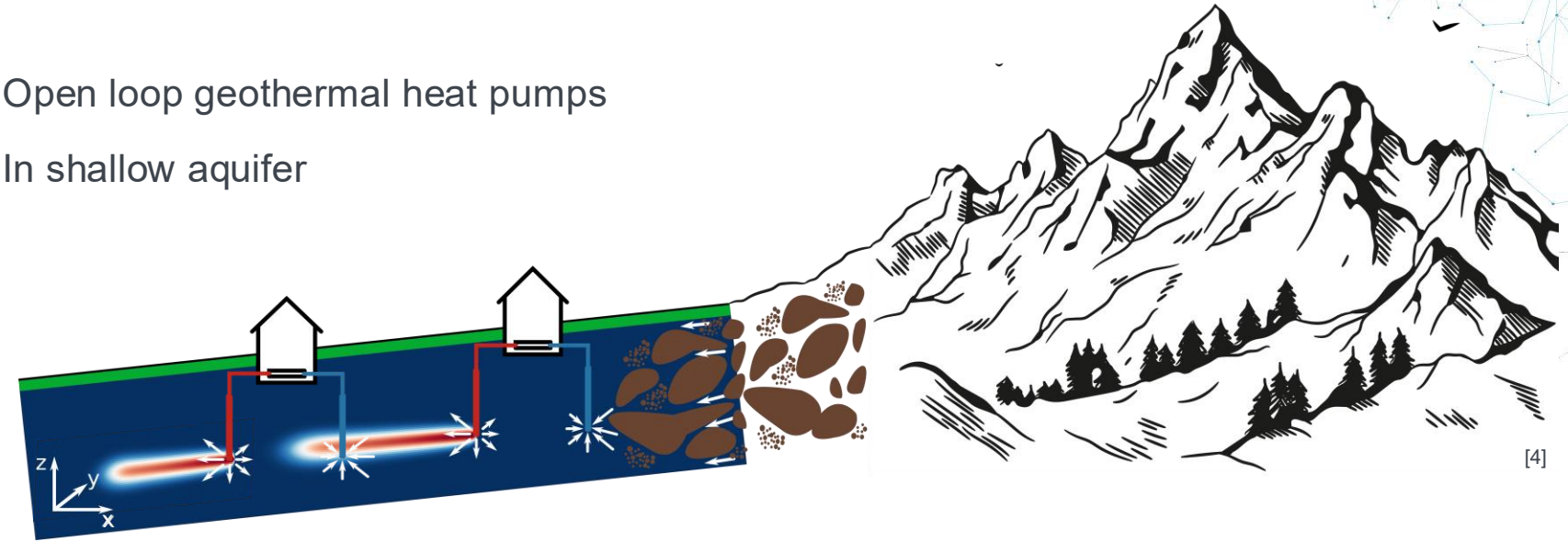
Geoscientific Context

- Open loop geothermal heat pumps
- In shallow aquifer



[4]

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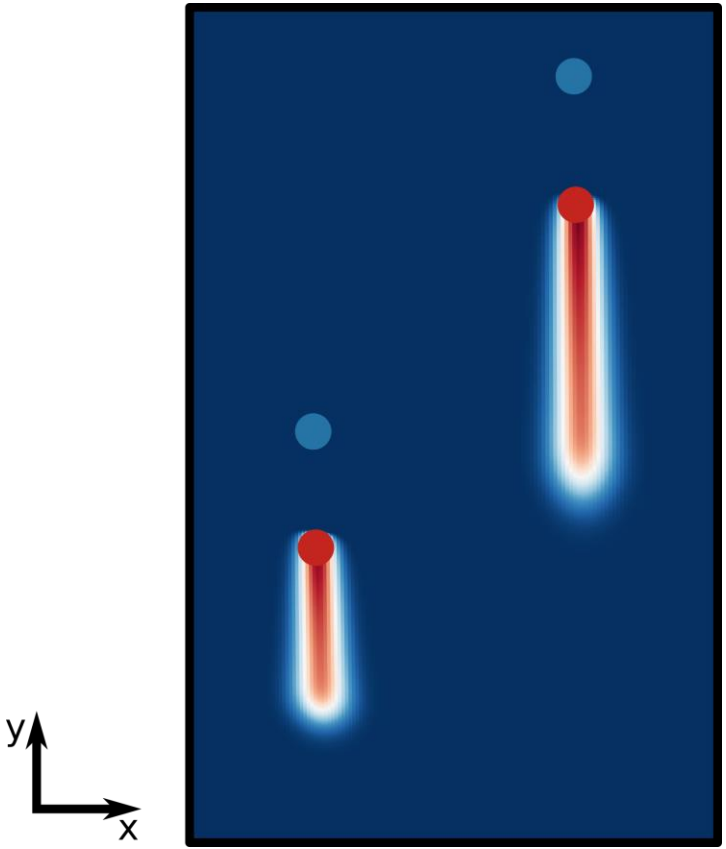

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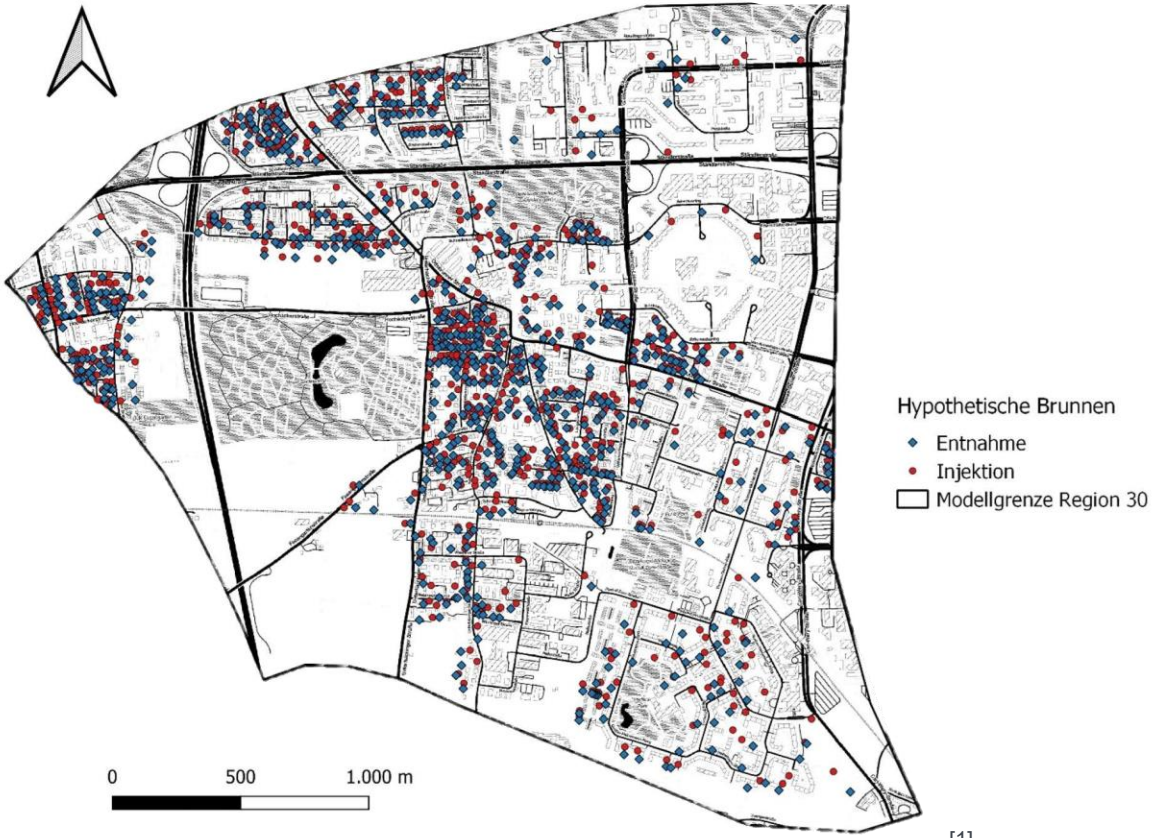
Hypothetische Brunnen
Entnahme
Injektion
Modellgrenze Region 30
0
500
1.000 m

[1]

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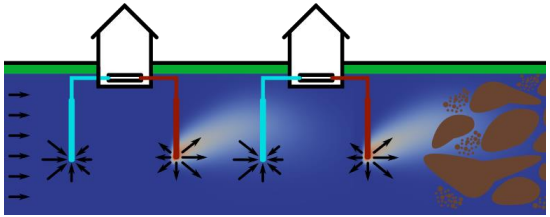
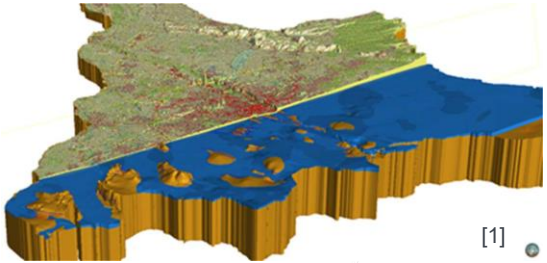
Classical modeling approaches

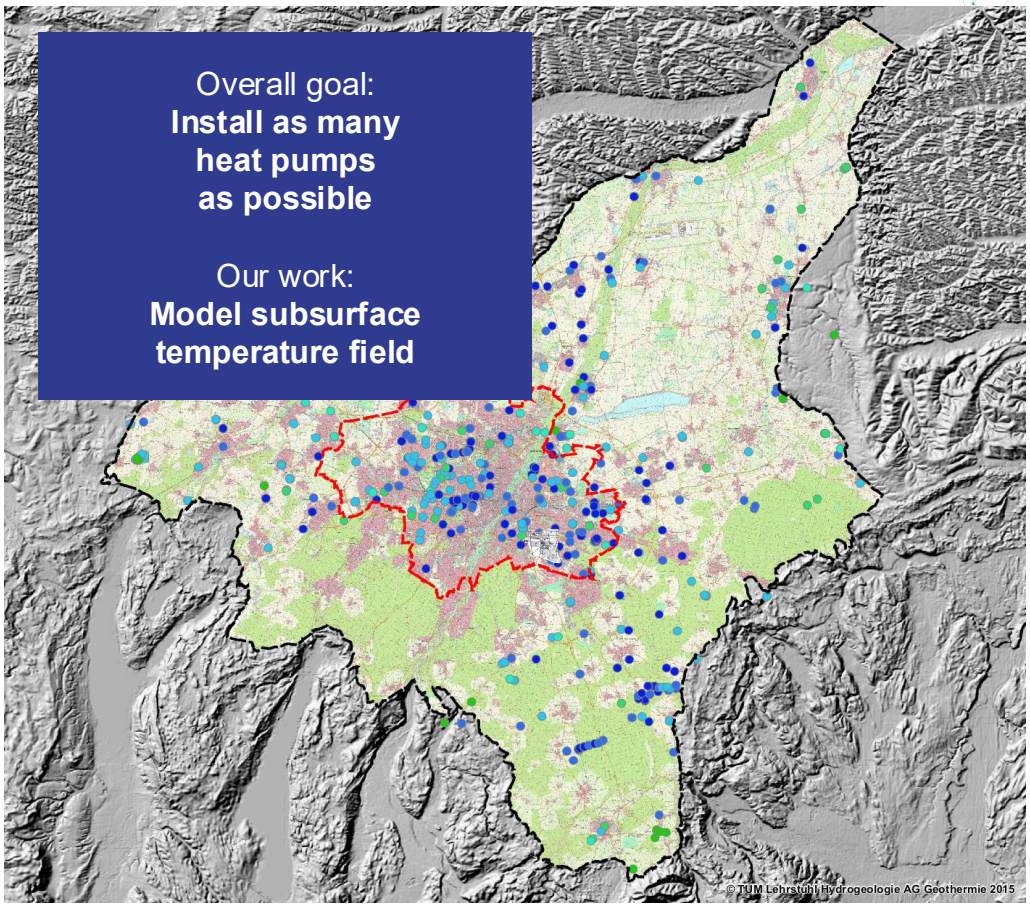


- Simulation, e.g., Pflotran
  - Slow, but accurate
- Analytical, e.g., LAHM
  - Fast, but limited use

⟶ Machine learning?
  - Fast, accuracy goal 0.1°C



[1]



Overall goal:
Install as many heat pumps as possible

Our work:
Model subsurface temperature field

[2]

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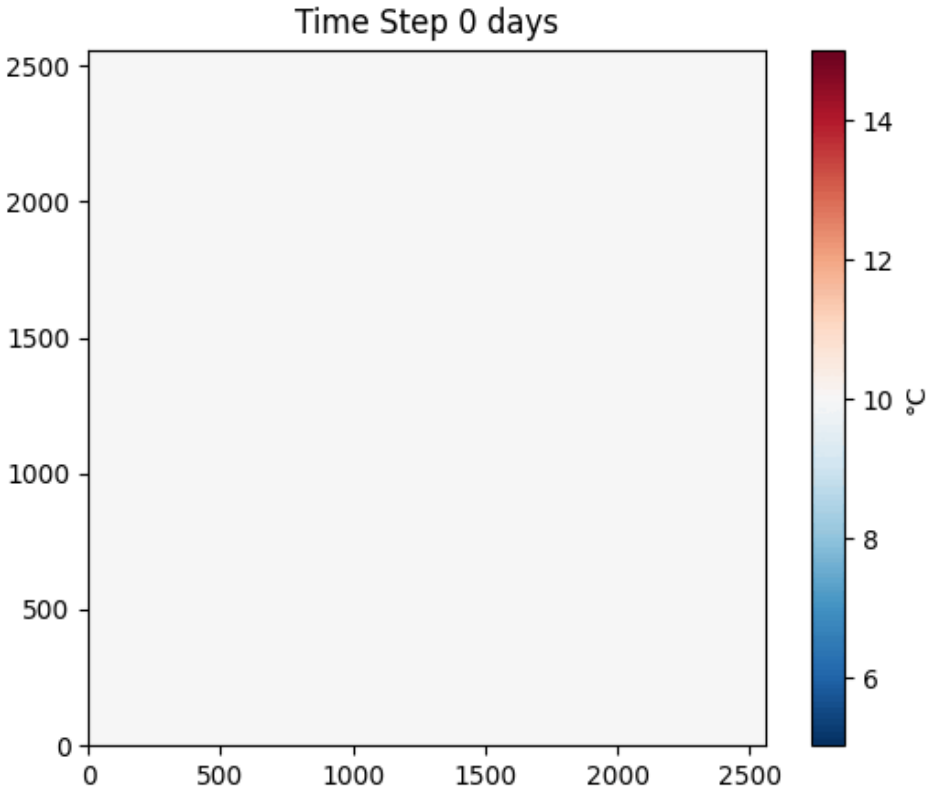
Benchmark Goal

- Large number of heat sources
- Heterogeneous subsurface parameters
- Timeseries or steady-state
- 2D, regular grid

- Methodological focus of the benchmark
  - Data scarcity
  - Scalability



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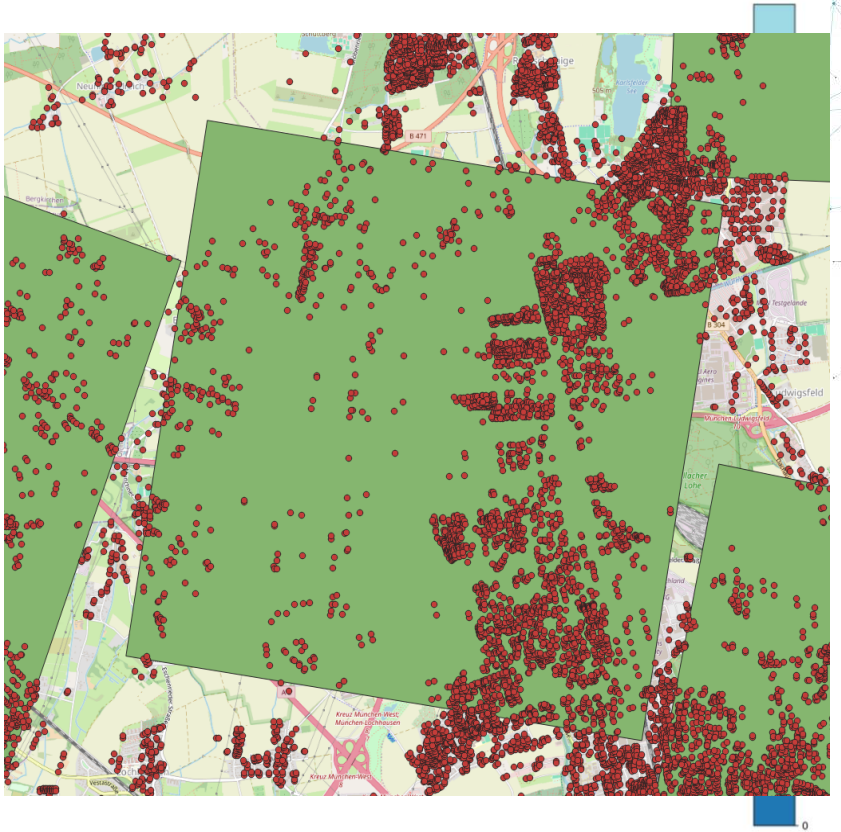
Data Generation

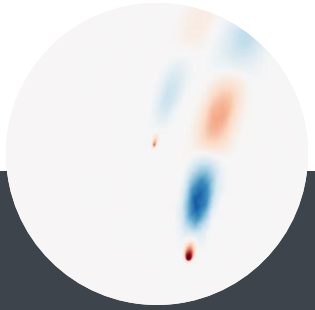
- Subsurface parameters
  - measurements from boreholes --> interpolationed maps [1] -> cut outs
- Heat pumps
  - Realistic positions [1]
  - Flow rates acc. to available groundwater
  - Real seasonal heating/ cooling cycles
- Temperature field (label)
  - Simulations with Feflow [6]
  - Preprocessing with Python



Stepwise Benchmark
Datasets

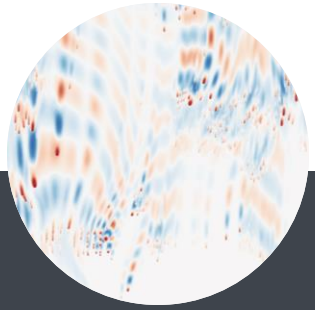
1. Single Heat Plume

- Lots of data (~ 1000)
- No interaction
- 1 injection, 1 extraction well
- 320x1280 cells



2. Two Interacting Plumes

- Less data (~ 400)
- Added interaction
- 2 injection, 2 extraction wells
- 1280x1280 cells



3. All Potential Pumps

- Scarce data (~ 20)
- Long-distance interactions
- All potential pumps active
- 2560x2560 cells

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Metrics

- Machine learning metrics:
  - MAE / L1
  - RMSE / squared L2
  - Maximum error
- Application specific metrics:
  - Accuracy threshold: 0.1°C

  --> Percentage above threshold (PAT 0.1)
  - Number of model parameters

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Benchmark Setup

Training

DATA

Step 1: Single Heat Plume Draft Unpublished Waiting for Feedback from Author

Apr 20, 2026

Pelzer, Julia; Böttcher, Fabian, 2026, "Step 1: Single Heat Plume", https://doi.org/10.18419/DARUS-5806, DaRUS, DRAFT VERSION

This dataset serves as training data for modeling the temperature field emanating from open-loop groundwater heat pumps. The dataset was simulated in 2D with Feflow using cut-outs from interpolated hydrogeological measurements of the Munich, Germany, region. Heat pump locations are chosen based on realistic positions, and extraction rates are adapt...

Step 2: Two Interacting Heat Plumes Draft Unpublished Waiting for Feedback from Author

Apr 20, 2026

Pelzer, Julia; Böttcher, Fabian, 2026, "Step 2: Two Interacting Heat Plumes", https://doi.org/10.18419/DARUS-5807, DaRUS, DRAFT VERSION

This dataset serves as training data for modeling the temperature field emanating from open-loop groundwater heat pumps. The dataset was simulated in 2D with Feflow using cut-outs from interpolated hydrogeological measurements of the Munich, Germany, region. Heat pump locations are chosen based on realistic positions, and extraction rates are adapt...

Step 3: Scaled-up Domain and Interactions of Heat Plumes Draft Unpublished Waiting for Feedback from Author

Apr 20, 2026

Pelzer, Julia; Böttcher, Fabian, 2026, "Step 3: Scaled-up Domain and Interactions of Heat Plumes", https://doi.org/10.18419/DARUS-5808, DaRUS, DRAFT VERSION

This dataset serves as training data for modeling the temperature field emanating from open-loop groundwater heat pumps. The dataset was simulated in 2D with Feflow using cut-outs from interpolated hydrogeological measurements of the Munich, Germany, region. Heat pump locations are realistic positions, and extraction rates are adapted to fit the av...

Raw Simulation Data of All Steps Draft Unpublished

Apr 17, 2026

Pelzer, Julia, 2026, "Raw Simulation Data of All Steps", https://doi.org/10.18419/DARUS-5920, DaRUS, DRAFT VERSION

This dataset contains unprocessed simulation data of modeling heat flow from open-loop groundwater heat pumps. The dataset was simulated in 2D with Feflow using cut-outs from interpolated hydrogeological measurements of the Munich, Germany, region. Heat pump locations are chosen based on realistic positions, and extraction rates are adapted to fit...

CODE

Stepwise-Benchmarking-HPs Public

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main | 1 Branch | 0 Tags | Go to file | Add file | Code

JuliaPelzer update readme — cba6425 · 15 hours ago — 4 Commits

code	minor: rm TODOs	15 hours ago
data/stepDummy	initial commit, including dummy data	15 hours ago
.gitignore	add gitignore	15 hours ago
README.md	update readme	15 hours ago
main.py	initial commit, including dummy data	15 hours ago
requirements.txt	initial commit, including dummy data	15 hours ago

README

Stepwise Benchmarking on Groundwater Heat Pump Modeling

This repository contains a very basic code framework to train and evaluate a model on the three separate steps of the `stepwise benchmark for groundwater heat pump modeling`. The code is structured in a way that allows you to easily modify the model architecture, training procedure, and input features. The data is already preprocessed; the dataloader contains a normalization function, so you can directly use it for training and

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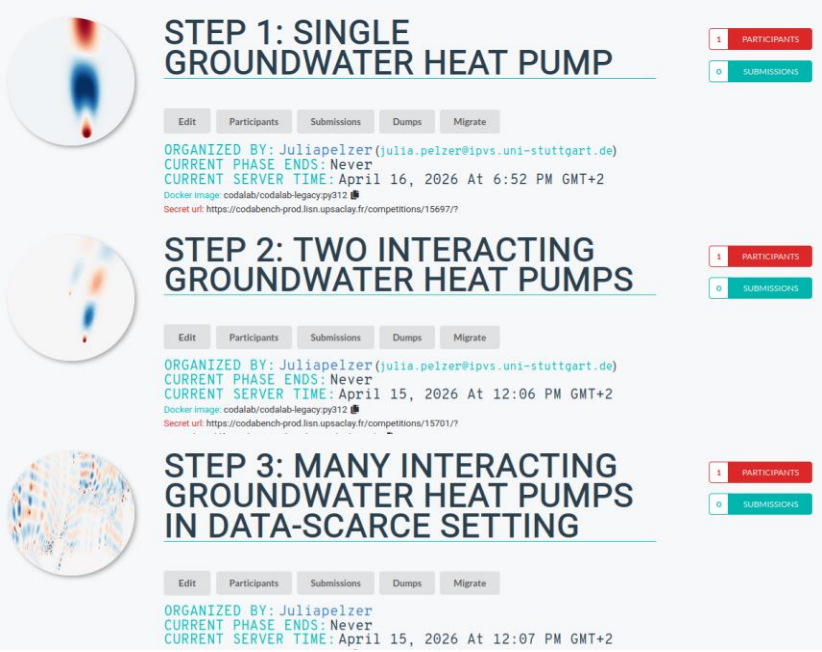
Benchmark Setup

Testing

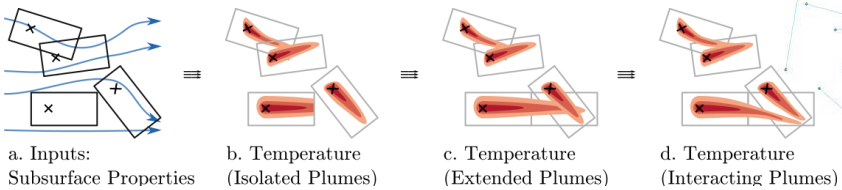
CODABENCH

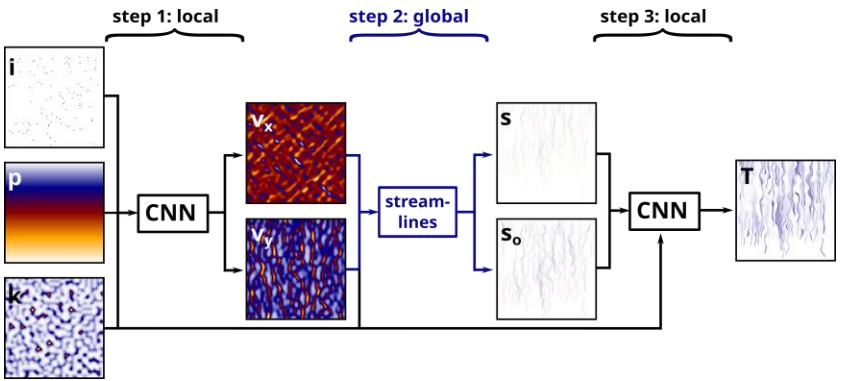
- Hidden test data
- Same metrics
- Easy submission



Prior SciML Work

- Based on simpler datasets (modeled with Pflotran)
- Hybrid, physics-inspired Convolutional Neural Network (CNN)-based approaches
  1. Flow-oriented boxes for UNet [PelzerSchulte24]
  2. Only model interaction based on step 1 [PelzerSchulte24]
  3. Split by physical process (advection vs diffusion) [PelzerEtAl26]





Summary

Status and Open Questions

- Training and validation data and code done
- Ready to test your (ML) approaches

- Open questions:
  - Additional metrics?
  - Experiences with platforms for testing data?

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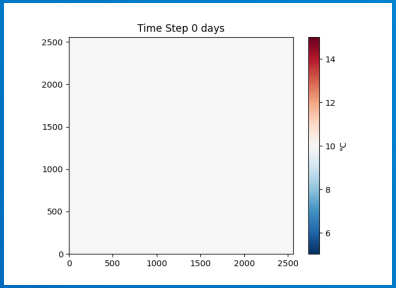
Thank you!



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Bibliography

[1] https://mediatum.ub.tum.de/doc/1692003/st6o6qfqfh21tcb6vyl88vegd.Schlussbericht_GEOKW_final_TIB_20221121.pdf (13/08/2024, goal)

[2] https://www.google.com/maps/place/M%C3%BCnchen/@48.1711074,11.4890278,11z/data=!4m6!3m5!1s0x479e75f9a38c5fd9:0x10cb84a7db1987d!8m2!3d48.1351253!4d11.5819805!16s%2Fm%2F02h6_6p?entry=ttu (13/08/2024, Munich map)
https://www.lfu.bayern.de/geologie/gepo/index.htm (13/03/2025)

[4] https://de.freepik.com/vektoren-kostenlos/monochrome-handgezeichnete-bergumrissillustration_41155991.htm#fromView=keyword&page=1&position=8&uuid=af8e0d07-dfa2-409d-b62b-a61d6c4618fa&query=Zugspitze+Alpen (12/03/25, Berg)

[5] https://commons.wikimedia.org/wiki/File:M%C3%BCnchner_Schotterebene.png?uselang=de (12/03/25)

[6] www.dhigroup.com/technologies/mikepoweredbydhi/feflow

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QR Codes

- Darus
- Github code
- Codabench challenge

data

code

codabench

step 1

step 2

step 3